LC-MS/MS Method Analysis

LC-MS/MS data were acquired on a high resolution Thermo Q Exactive HF mass spectrometer with Thermo Vanquish UHPLC+ Focused in both positive and negative electrospray ionization. Standards were analyzed using a Phenomenex Kinetex PFP (1.7 μm, 2.1 mm X 30 mm) with a SecurityGuard™ ULTRA Cartridges UHPLC PFP (2.1mm). The mobile phases were H_2O with 0.1% formic acid (A) and ACN with 0.1% formic acid (B). Gradient elution was performed at a flow rate of 0.4 mL/min under the following program: from 0–4.50 min % B changed linearly from 7%–30%, changed to 100% B from 4.50-6.50 min, then held at 100% B for 1.00 min, returned to 7% B over the next 0.50 min, and held at 7% B for 2 min. The column temperature was kept at 45°C.

Mass spectral data on the Q Exactive HF was collected in positive and negative ionization mode using three Full MS with data dependent- MS^2 scan events. Each event was conducted using three separate normalized collision energies (NCE): 35, 45, 65. In general, Full MS was acquired with the resolution at 60,000, the AGC target set to 1×10^6, the maximum injection time set to 100 ms, and the data acquired in centroid. In general, dd-MS^2 was acquired with the resolution at 15,000, with the AGC target 5×10^5, maximum injection time set to 50 ms, the loop count set to 5, TopN set to 5, the isolation window of 1.0 m/z, dynamic exclusion set to 0.8 s, and data acquired in centroid. A inclusion list was used to target MS^2.